

Investigating a RAT Infection Using Wireshark: A Step-by-Step Analysis

Introduction

In this investigation, we analyse a packet capture (PCAP) file to identify a potentially compromised Windows system communicating with a known Remote Access Trojan (RAT). Using Wireshark, we systematically uncover key indicators such as the infected host's IP address, MAC address, hostname, and associated user details.

The known malicious indicator used for this analysis is:

- RAT IP Address: **45.131.214[.]85**
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Identifying the Infected Windows Client

To begin the investigation, the PCAP file was loaded into Wireshark and filtered using:

"ip.addr == 45.131.214.85"

This filter isolates all traffic where the RAT IP appears as either the source or destination. As a result, the dataset was reduced from 15,512 packets to 550 relevant packets.

Further analysis revealed that a single internal host, 10.2.28[.]88, was consistently communicating with the malicious IP.

To validate this, the Statistics → Conversations feature in Wireshark was used. The results confirmed that 100% of the filtered traffic involved communication exclusively between 10.2.28[.]88 and 45.131.214[.]85, strongly indicating that this internal host is the infected system.

- Infected Windows Client IP: **10.2.28[.]88**

Ethernet · 1		IPv4 · 1	IPv6	TCP · 1	UDP					
Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A
10.2.28.88	45.131.214.85	550	96 kB	25	550	100.00%	276	79 kB	274	17 kB

Determining the MAC Address

Using the same filtered dataset, the first packet was examined in detail. Within the Ethernet frame header, the source MAC address corresponding to the infected system was identified.

- MAC Address: **00:19:d1:b2:4d:ad**

The image shows a Wireshark packet capture analysis. The top pane displays a list of filtered packets (No. 2569 to 4134) between source 10.2.28.88 and destination 45.131.214.85. The middle pane shows the details of packet 2569, highlighting the Ethernet II frame header. The source MAC address is identified as 00:19:d1:b2:4d:ad. The bottom pane shows the raw packet data in hexadecimal and ASCII.

Frame 2569: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0

Ethernet II, Src: Intel_b2:4d:ad (00:19:d1:b2:4d:ad), Dst: Cisco_41:ae:25 (00:05:00:41:ae:25)

Destination: Cisco_41:ae:25 (00:05:00:41:ae:25)

Source: Intel_b2:4d:ad (00:19:d1:b2:4d:ad)

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 10.2.28.88, Dst: 45.131.214.85

Transmission Control Protocol, Src Port: 51912, Dst Port: 443, Seq: 0, Len: 0

Extracting the Hostname

To identify the system hostname, DHCP traffic was analysed using the filter:

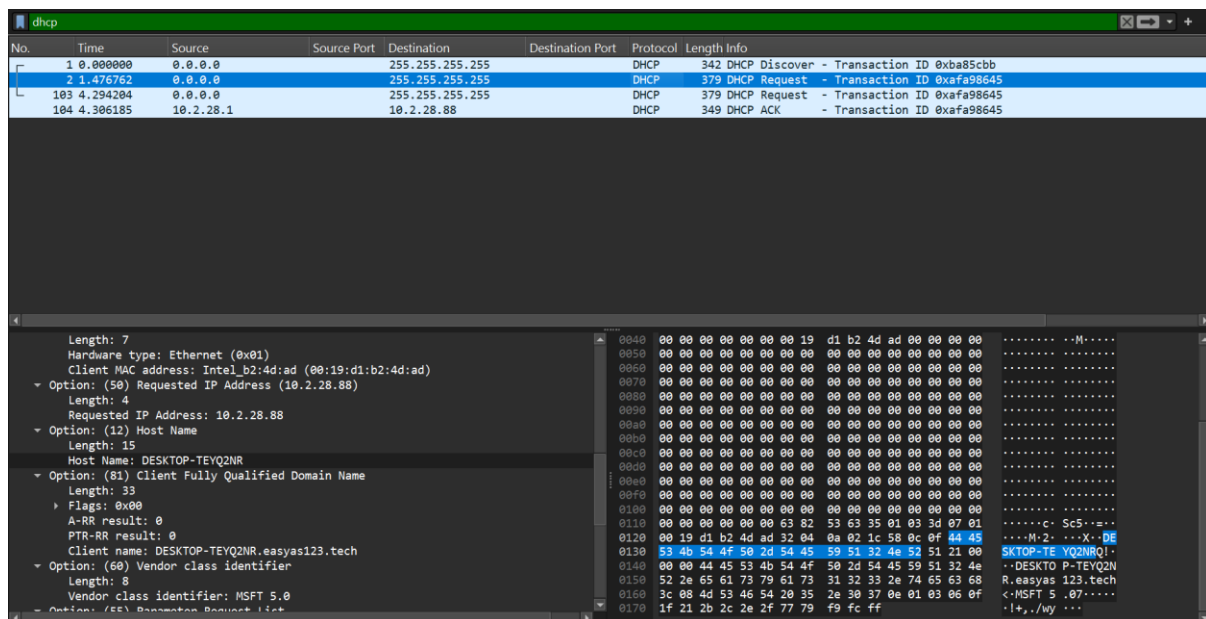
“dhcp”

DHCP Request packets often include the hostname of the client as part of the request.

Among the filtered results, four DHCP packets were observed.

By correlating the MAC address (00:19:d1:b2:4d:ad) with the DHCP Request packet, the hostname associated with the infected system was identified as:

- Hostname: **DESKTOP-TEYQ2NR**



No.	Time	Source	Source Port	Destination	Destination Port	Protocol	Length	Info
1	0.000000	0.0.0.0		255.255.255.255		DHCP	342	DHCP Discover - Transaction ID 0xba85cbb
2	1.476762	0.0.0.0		255.255.255.255		DHCP	379	DHCP Request - Transaction ID 0xafa98645
103	4.294294	0.0.0.0		255.255.255.255		DHCP	379	DHCP Request - Transaction ID 0xafa98645
104	4.386185	10.2.28.1		10.2.28.88		DHCP	349	DHCP ACK - Transaction ID 0xafa98645

Length	Info	Raw
7	Hardware type: Ethernet (0x01)	0040 00 00 00 00 00 00 19 d1 b2 4d ad 00 00 00 00
	Client MAC address: Intel b2:4d:ad (00:19:d1:b2:4d:ad)	0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Option: (50) Requested IP Address (10.2.28.88)		0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Length: 4		0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Requested IP Address: 10.2.28.88		0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Option: (12) Host Name		0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Length: 15		00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Host Name: DESKTOP-TEYQ2NR		00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Option: (81) Client Fully Qualified Domain Name		00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Length: 33		00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Flags: 0x00		00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
A-RR result: 0		00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
PTR-RR result: 0		0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
Client name: DESKTOP-TEYQ2NR.easys123.tech		0110 00 00 00 00 00 00 00 63 82 53 63 35 01 03 3d 07 01
Option: (60) Vendor class identifier		0120 00 19 d1 b2 4d ad 32 04 0a 02 1c 58 0c 0f 44 45
Length: 8		0130 53 4b 54 4f 50 2d 54 45 59 51 32 4e 52 51 21 00
Vendor class identifier: MSFT 5.0		0140 00 00 44 45 53 4b 54 4f 50 2d 54 45 59 51 32 4e
Option: (55) Reservation Request List		0150 52 2e 65 61 73 79 61 73 31 32 33 2e 74 65 63 68
		0160 3c 09 4d 53 46 54 20 35 2e 30 37 0e 01 03 06 0f
		0170 1f 21 2b 2c 2e 2f 77 79 f9 fc ff

Identifying the User Account

Next, Kerberos authentication traffic was analyzed to determine the user account associated with the infected host. Applying the filter:

“kerberos”

allowed isolation of authentication-related packets.

The field `kerberos.CNameString` was used to extract the client’s principal name from the Kerberos exchange. This revealed the user account:

- User Account: **brolf**

Wireshark packet capture showing Kerberos authentication traffic. The packet list shows a TGS-REP packet (No. 253) from 10.2.28.88 to 10.2.28.88. The packet details pane shows the 'CNameString' field with the value 'brolf'. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Source Port	Destination	Destination Port	Protocol	Length	Info
243	16.565177	10.2.28.88	59985	10.2.28.2	88	KRBS	270	AS-REQ
251	16.578763	10.2.28.88	59986	10.2.28.2	88	KRBS	350	AS-REQ
253	16.572761	10.2.28.2	88	10.2.28.88	59986	KRBS	258	AS-REP
265	16.576639	10.2.28.2	88	10.2.28.88	59987	KRBS	199	TGS-REP
288	16.745555	10.2.28.2	88	10.2.28.88	59989	KRBS	333	TGS-REP
300	16.747703	10.2.28.2	88	10.2.28.88	59990	KRBS	154	TGS-REP
369	16.813165	10.2.28.2	88	10.2.28.88	59992	KRBS	303	TGS-REP
400	16.848884	10.2.28.2	88	10.2.28.88	59995	KRBS	349	TGS-REP
2672	52.079058	10.2.28.2	88	10.2.28.88	51914	KRBS	333	TGS-REP

Packet (258 bytes) Reassembled TCP (1664 bytes)

CNameString (kerberos.CNameString), 5 bytes

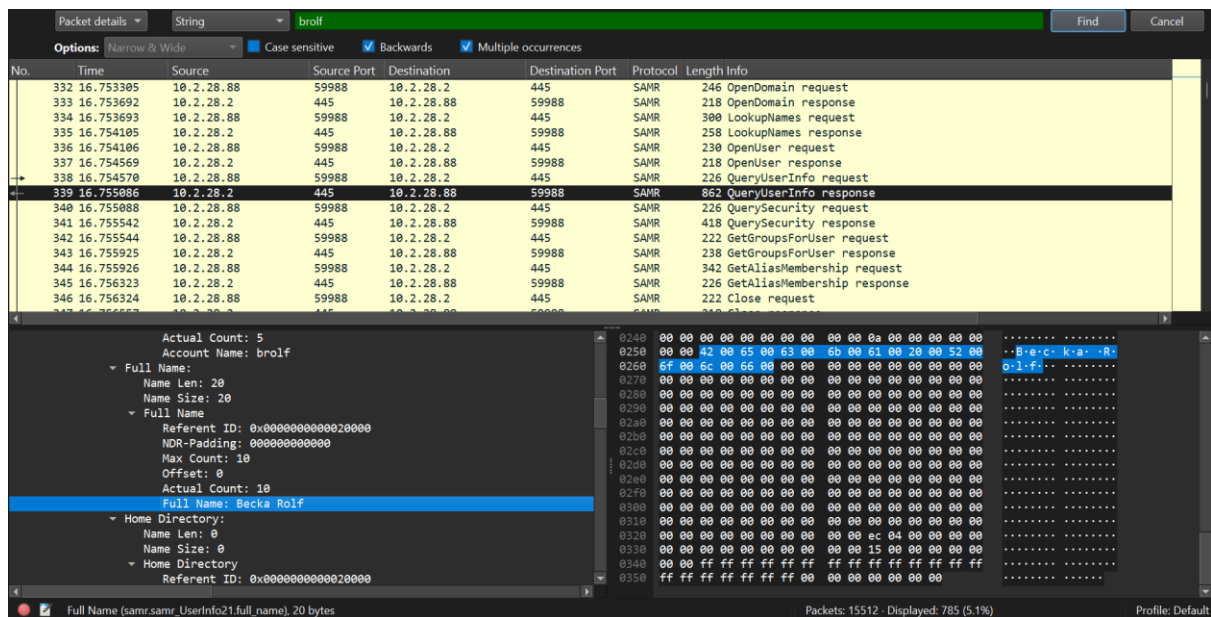
Packets: 15512 · Displayed: 9 (0.1%) Profile: Default

Determining the Full Name of the User

To obtain more details about the user, the Find Packet feature in Wireshark was used to search for occurrences of the username “brolf.”

Among the results, packet number 339 contained additional user information. Inspecting this packet revealed a field labelled Full Name, which identified the user as:

- Full Name: **Becka Rolf**



Summary of Findings

Attribute	Value
Infected IP Address	10.2.28[.]88
MAC Address	00:19:d1:b2:4d:ad
Hostname	DESKTOP-TEYQ2NR
User Account	brolf
Full Name	Becka Rolf

Conclusion

Through systematic analysis of network traffic using Wireshark, we successfully identified the compromised Windows system and associated user details. By leveraging protocol-specific filters such as IP, DHCP, and Kerberos, we were able to pivot from a known malicious IP to uncover critical host-level indicators.

This investigation highlights the importance of network visibility and structured analysis techniques in incident response and threat hunting.
